TECHNICAL DOSSIER

Hirobot

Friendly educational tank robot for learning robotics and different aspects of programming.

DOSSIER DATA

PROJECT LEAD Claudia Élez Mencía	REPOSITORY github.com/RoboTech-URJC/HiroBot
STATUS In development	LAST UPDATE June 2026

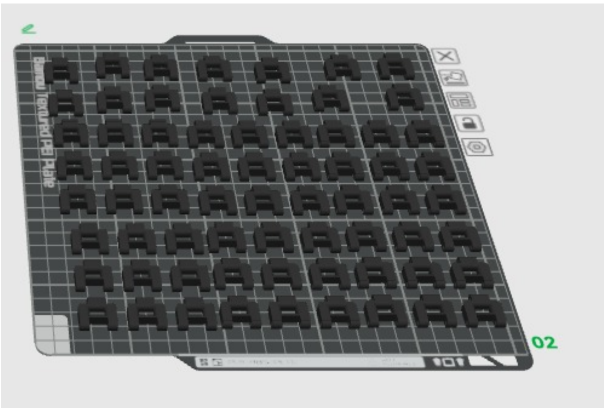
Project Description

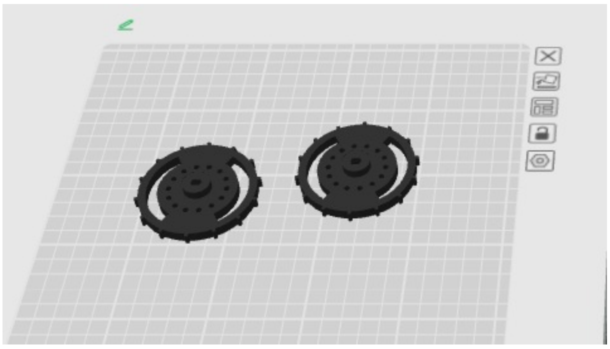
The Hirobot project consists of building a robot whose goal is to serve as an educational platform for learning robotics and different aspects of programming. For this reason, the goal is to develop a tank robot with a friendly appearance.

Development During the 2025–2026 Year

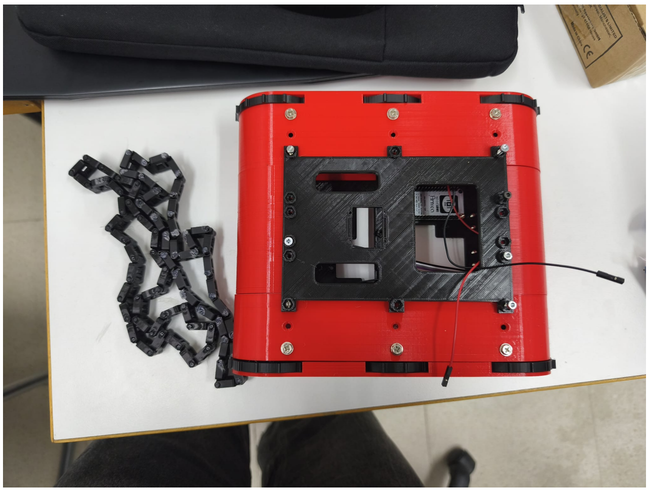
This project was started this year, although it had been organized to begin during the previous academic year. Therefore, a large part of the materials had already been purchased and were available in the association workspace.

During the first months, work began by printing the main part of the tank, the base, which is based on the following project: thingiverse.com/thing:2039843.

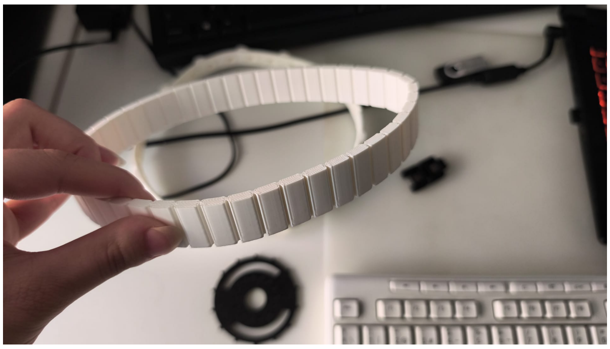




Once the full base had been printed, a basic tank movement program was tested with the micromotors available in the workspace. The team concluded that the track type should be changed, four motors should be used instead of two, and these new motors should be purchased together with the L298N module.



Tracks: the first approach used a PLA printed chain track, but the results were not as expected. Throughout the year, different track designs have been designed and printed in TPU.



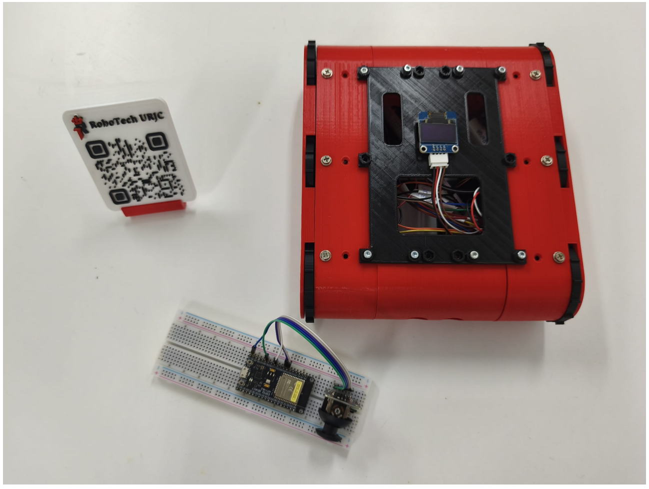
Work is still ongoing on the track design, with a nearly final version pending testing over the summer.

At the same time, MQTT communication began to be implemented using a Mosquitto broker, enabling communication with another ESP82 board, a remote controller, and a visual PC interface through Node-RED. From this interface, the tank can also be remote-controlled and future implementations can be developed.

Current Status and Future Lines

The tank base is assembled except for the tracks, and MQTT communication has been achieved both with another ESP82 board and with the Node-RED interface.

The project has been lent out over the summer with the intention of finalizing the tank base design, working on tank movement, and also focusing on the robot's programming.



Future goals include:

- Implementing a camera for recognizing people and the environment. Initially, this will be tested with a PiCamera and a Raspberry Pi.
- Implementing a microphone and speaker for communication with users.
- Beginning the design of the robot's neck / head.

Other Images or Resources

This year's material can be accessed through this Drive folder: Hirobot 2025-2026.